

data iteratively. For example, when the `find()` method is used to find the documents present in the given collection, it returns a pointer that points to the documents of the collection. This pointer is known as a cursor. Queries return a database cursor, which lazily returns the batches of documents as needed.

There are a lot of meta operations one can perform on a cursor, including skipping a certain number of results, limiting the number of results returned, and sorting the results.

Indexes

Indexes support the efficient execution of queries in MongoDB. Without indexes, MongoDB must scan every document in a collection to select those documents that match the query statement. If an appropriate index exists for a query, MongoDB can use the index to limit the number of documents it must inspect.

The index is a special data structure, B-Tree, which stores the value of a specific field or a set of fields. The ordering of the index entries supports efficient equality matches and range-based query operations. Using B-Tree indexes significantly reduces the number of comparisons to find the document. The index types are:

Single field: By default, the `_id` field is indexed. Additionally, any field from the document can be indexed. This increases the performance of the query for the select operations done on the indexed field, such as sort by the field or search by it.

Compound index: This supports queries on multiple fields. The order of the indexed fields has a strong impact on the effectiveness of a particular index for a given query.

Multikey index: Multikey index is used to index for the content stored in an array. It creates separate index entries for each value in the array.

Geospatial index: This is the index on the geospatial data. It comprises two special indexes — 2d indexes that use planar geometry when returning results and 2dsphere indexes that use spherical geometry to return results.

Text index: Text index supports queries on the string content in a collection. We can have more than one string field for creating this index. The weight of an indexed field denotes the significance of the field relative to the other indexed fields in terms of the text search score. For example, in the code given below three fields are used to create the index, each with its own weights:

```
db.blog.createIndex(
{
  content: "text",
  keywords: "text",
  about: "text"
},
{

```

```
weights: {
  content: 10,
  keywords: 5
},
name: "TextIndex"
}
```

For each indexed field in the document, MongoDB multiplies the number of matches by the weight and sums the results. Using this sum, MongoDB calculates the score for the document.

Why you should use MongoDB

MongoDB offers flexibility in the data model and has quicker releases. It is easy to install and offers data scalability. A distributed system and cloud computing deliver data resiliency. MongoDB supports various data types with ease, such as time series, geospatial, polymorphic data, etc. It can help create rich, data-driven applications, and has balanced high performance reads (indexes) and writes. As data grows, sharding helps by horizontally scaling and saving data across multiple instances. It's cost-effective since it has an open source version.

The challenges

As MongoDB is distributed in nature, transactions are a challenge. However, starting from MongoDB 4.2, distributed transactions can be used across multiple operations, collections, databases, and documents. Joins are not customary; there is a stage called `$lookup` for joins that comes with high memory cost. Document data size is limited to 16MB, and a limited number of levels are available in nesting.

However, as MongoDB is still evolving, enough work is being done to address all these limitations. 

References

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The author is a lifelong learner influenced by the quote:
"The more I learn, the more I realise how much I don't know."
Albert Einstein